

Graph Class - 3

Special class

directed
→ bfs → π

→ Topological Sort

Math → English

English → Sanskrit

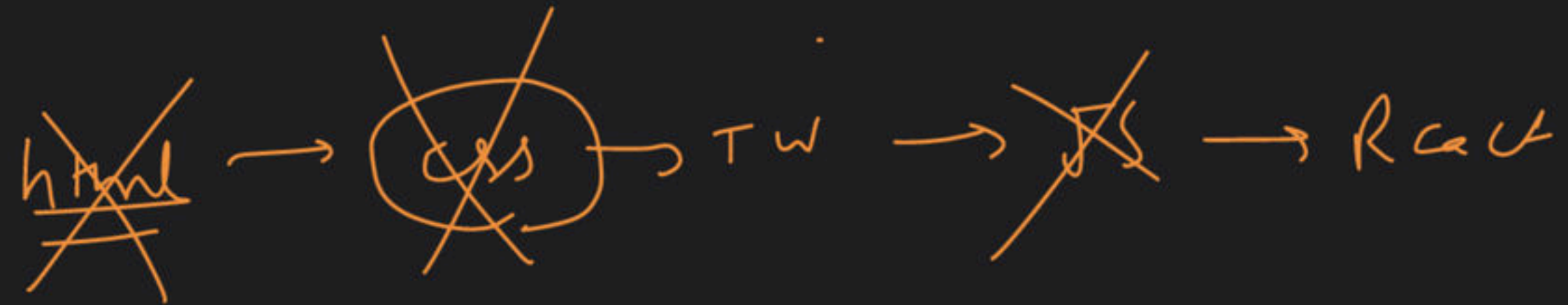
Sanskrit → Hindi

Hindi → Urdu

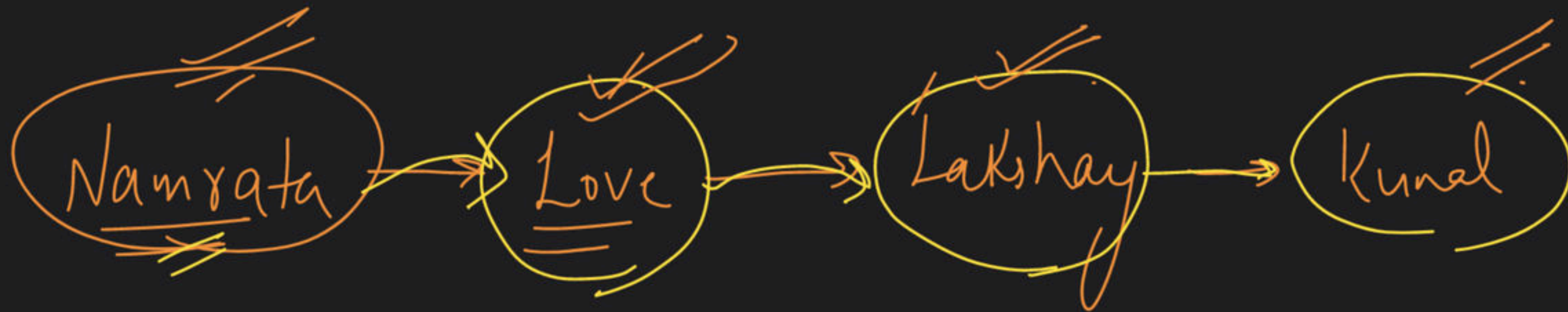
flow

Hindi → P → S → C → M

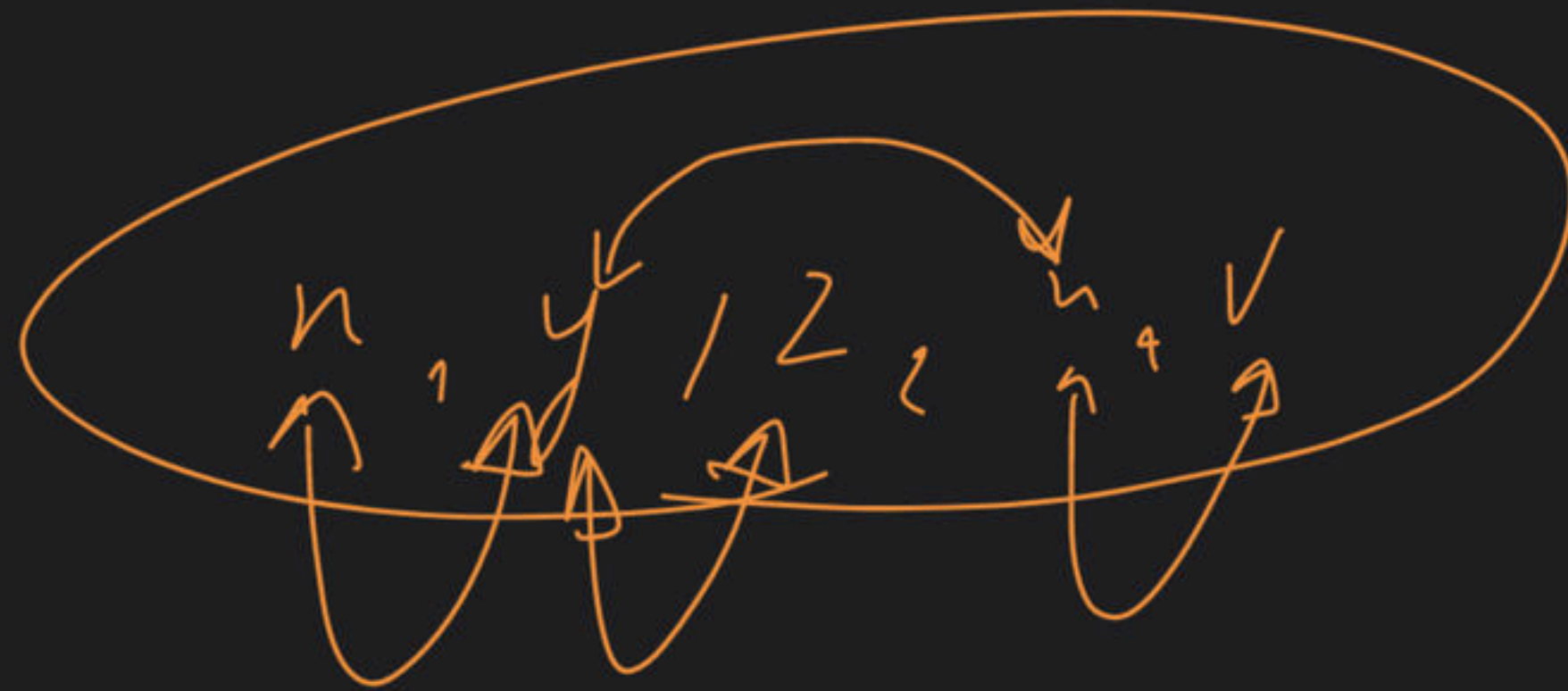
web-dev



Notes



→ Topological Sort { DAG }
 directed acyclic graph
 Linear Ordering + (rule)



Edge List

n → y

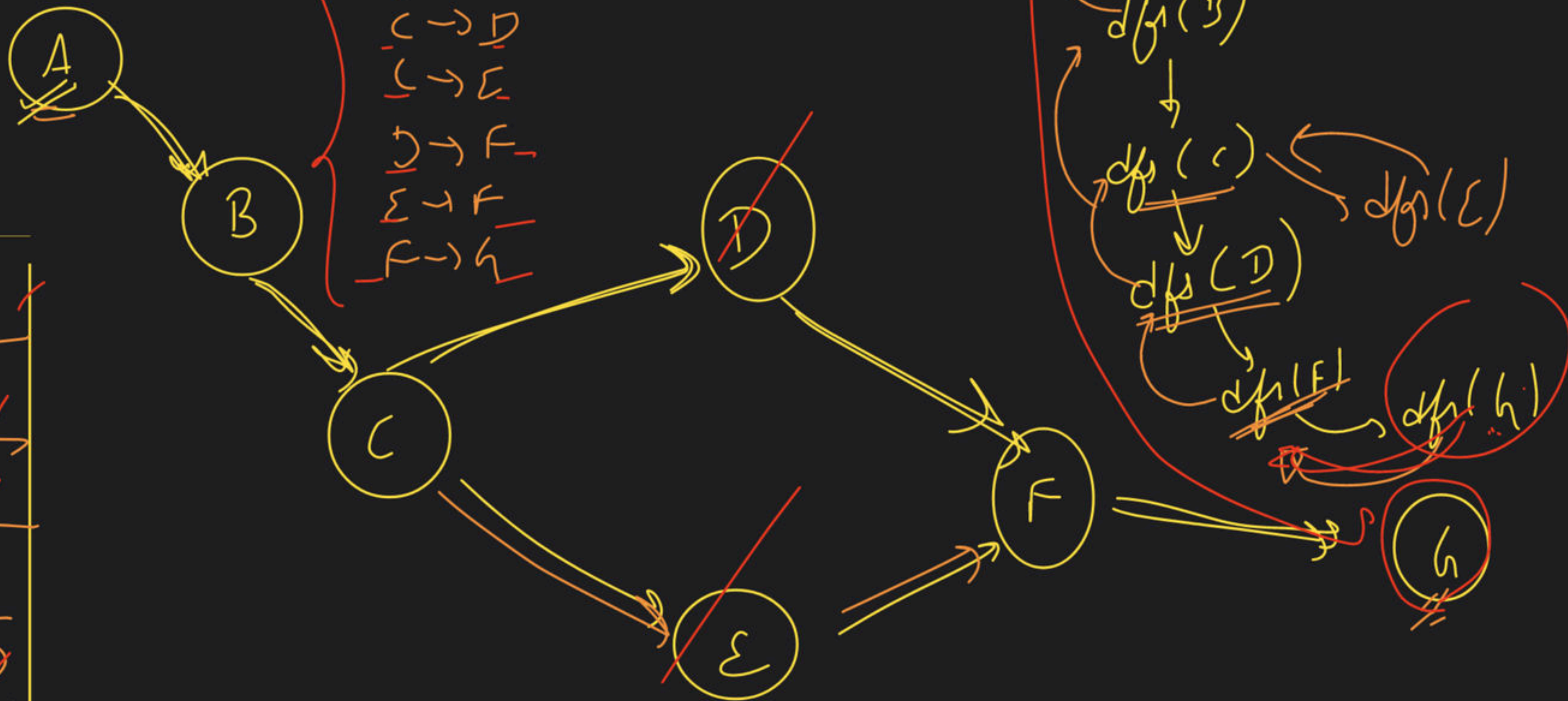
y → 1

y → 2

1 → u

2 → u

u → v



A	/
B	/
C	/
E	/
D	/
F	/
G	/

Stack

A -> B -> C -> E -> D -> F -> G

→ Topological sort → BFS initial state



src

0

queue

1

2

3

4

5

6

indegree

0	→ 0
1	→ 1
2	→ 1
3	→ 1
4	→ 2
5	→ 1
6	→ 1

map

① indegree calculated

② traverse indegree

= 0

q → push

③ main BFS logic

disconnected component

?

indegree

A → D

B → 1 }

C → 1 }

D → ~~2~~ 0 }

E → 0 }

F → 0 }

G → ~~2~~ 0

~~A~~ | ~~B~~ | ~~C~~ | ~~D~~ | ~~E~~ | ~~F~~ | ~~G~~

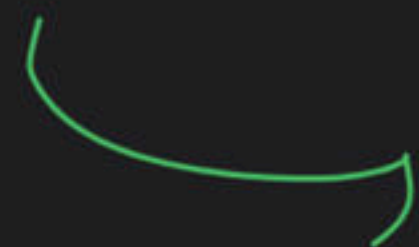
ans (I)

→

A	B	C	D	E	F	G
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Qy

Cycle detection



Directed graph

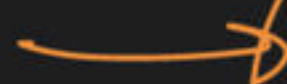


BFS?
=

DAG



TS



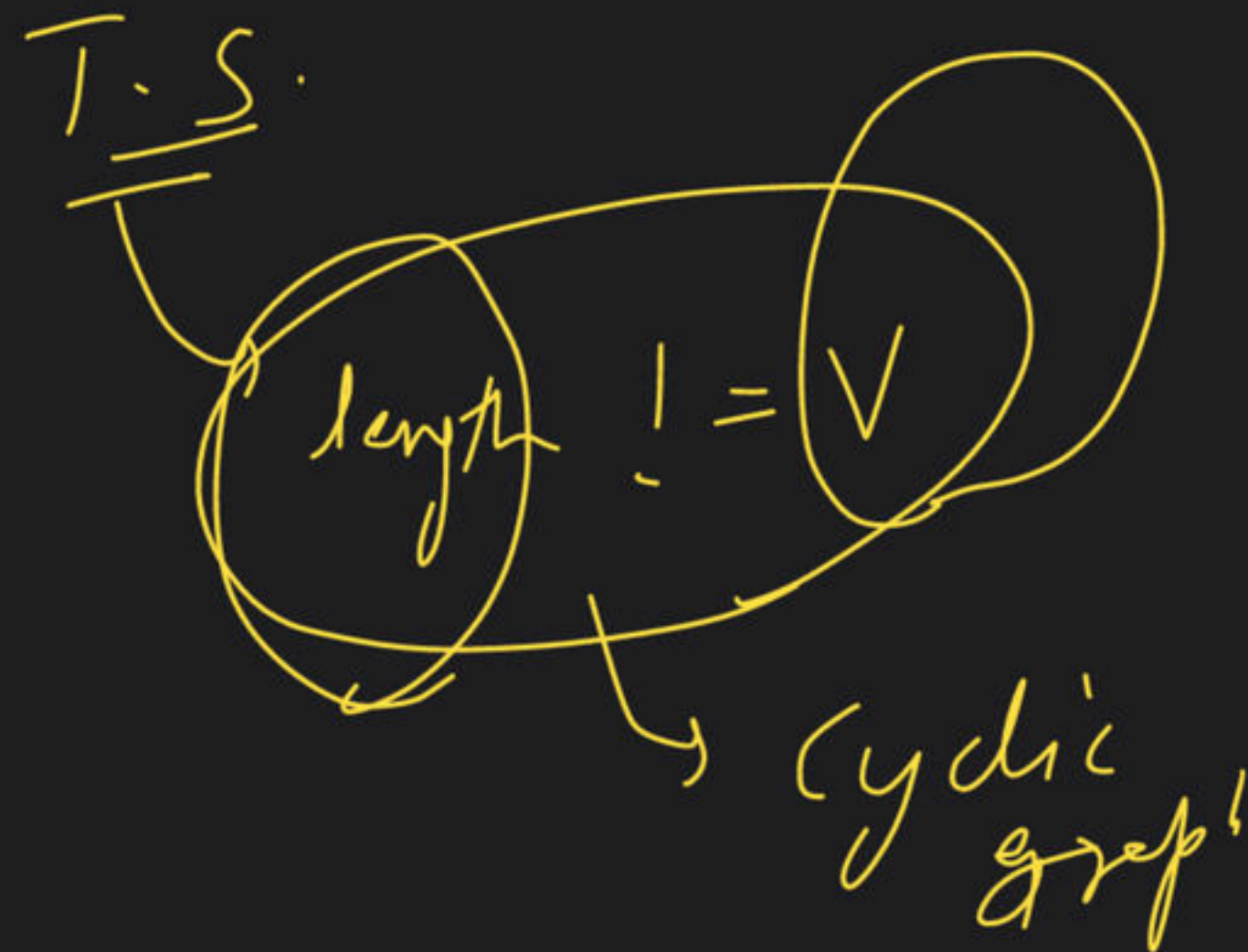
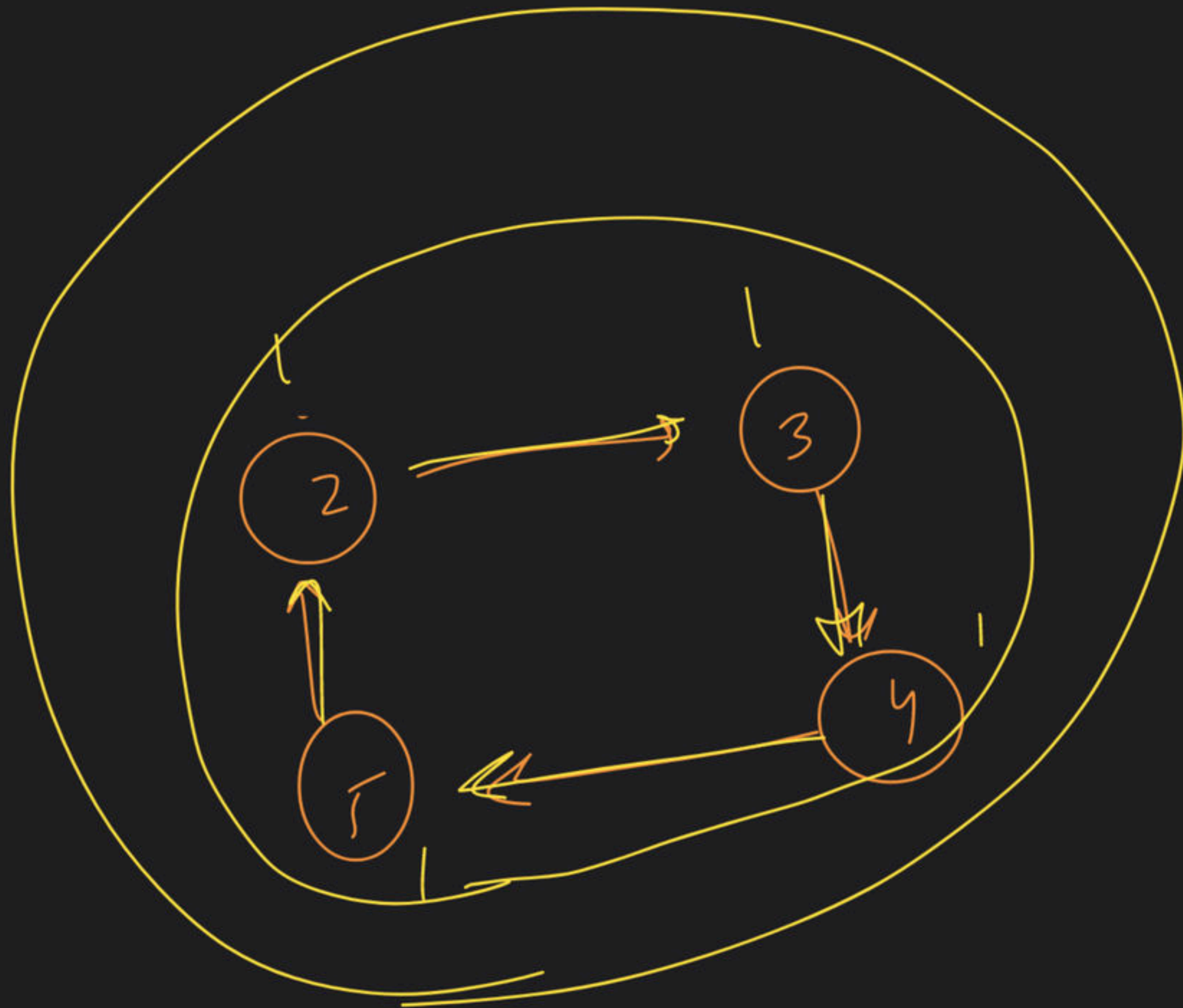
Directed
Acyclic graph



Valid
T-S



Cyclic graph → ~~TS~~



Kal

8:30 ↔ 9:00







































